Started on	Wednesday, 7 October 2026, 9:02 PM
State	Finished
Completed on	Wednesday, 7 October 2026, 9:32 PM
Time taken	29 mins 17 secs
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 10.00 out of 10.00

Playing with Numbers:

Ram and Sita are playing with numbers by giving puzzles to each other. Now it was Ram term, so he gave Sita a positive integer 'n' and two numbers 1 and 3. He asked her to find the possible ways by which the number n can be represented using 1 and 3.Write any efficient algorithm to find the possible ways.

Example 1:

Input: *6*

Output:*6*

Explanation: *There are 6 ways to 6 represent number with 1 and 3*

1+1+1+1+1+1

3+3

1+1+1+3

1+1+3+1

1+3+1+1

3+1+1+1

Input Format

First Line contains the number n

Output Format

Print: The number of possible ways 'n' can be represented using 1 and 3

Sample Input

6

Sample Output

6

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
long long int countWays(int n) {
    if (n<0)return 0;
    if (n==0 || n==1||n==2)return 1;

    long long int dp[n+1];

    dp[0]=1;
    dp[1]=1;
    dp[2]=1;

    for(int i=3;i<=n;i++){
        dp[i]=dp[i-1]+dp[i-3];

    }
    return dp[n];
}
int main(){
    int n;
    if(scanf("%d",&n)==1) {
        printf("%lld\n", countWays(n));
    }
    return 0;

}
```

	Input	Expected	Got	
✔	6	6	6	✔
✔	25	8641	8641	✔
✔	100	24382819596721629	24382819596721629	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.